SoccerGMR Ablation Study: 8 Backbones × 3 Configurations

Comparing Baseline GMR vs. GMR + DQ-CGPv3 vs. GMR + DQ-CGPv3 + GEQ across Standard Test (1,036 queries) and Full-Decodable Test (2,970 queries). G1-AUC = integral of G-mIoU@1 across tau in [0.00, 1.00].

		Standard Test Split (1,036 queries)						Full-Decodable Test Split (2,970 queries)					
		Loc	*Overall*	*Existence & Null-set Rejection*				*Loc*	*Overall*	*Existence & Null-set Rejection*			
Backbone	**Configuration**	**mAP (%)**	**G1-AUC (%)**	**Rej-F1@.5**	**Rej-F1@.4**	**Rej-F1@.6**	**AUROC (%)**	**mAP (%)**	**G1-AUC (%)**	**Rej-F1@.5**	**Rej-F1@.4**	**Rej-F1@.6**	**AUROC (%)**
Moment-DETR	1. GMR Baseline	6.68	29.04	67.94	0.00	74.94	70.20	5.47	27.01	72.08	0.00	77.46	78.91
	2. GMR + DQ-CGPv3	10.66	34.85	72.20	31.34	75.17	72.09	9.99	32.82	74.92	33.26	78.42	80.24
	3. GMR + DQ-CGPv3 + GEQ	**12.64**	**42.92**	**75.48**	**75.20**	**75.23**	**73.48**	**11.63**	**40.55**	**78.93**	**77.98**	**79.01**	**80.37**
EaTR	1. GMR Baseline	16.49	37.90	65.60	57.11	73.46	76.56	15.19	36.53	67.38	57.25	75.06	81.59
	2. GMR + DQ-CGPv3	16.02	35.82	65.27	45.07	73.67	**77.19**	15.35	34.99	65.29	47.29	74.33	80.38
	3. GMR + DQ-CGPv3 + GEQ	**17.48**	**44.39**	**75.43**	**74.46**	**75.23**	76.17	**15.67**	**42.50**	**78.73**	**77.32**	**78.44**	**83.04**
QD-DETR	1. GMR Baseline	14.62	38.72	71.48	61.73	73.29	**77.01**	12.65	36.93	72.49	62.28	76.02	81.75
	2. GMR + DQ-CGPv3	15.82	39.54	69.02	57.34	73.20	74.24	13.93	38.05	71.63	60.00	76.56	81.64
	3. GMR + DQ-CGPv3 + GEQ	**17.28**	**42.35**	**72.25**	**66.73**	**74.52**	75.18	**14.20**	**40.24**	**75.61**	**70.30**	**77.55**	**82.48**
CG-DETR	1. GMR Baseline	**18.65**	34.50	57.47	35.46	71.09	**76.40**	**14.67**	33.53	63.23	41.16	73.34	81.81
	2. GMR + DQ-CGPv3	13.49	36.84	75.04	57.47	**75.23**	74.63	12.49	35.40	76.97	61.20	77.01	79.33
	3. GMR + DQ-CGPv3 + GEQ	14.75	**43.76**	**75.31**	**75.58**	75.11	75.77	13.17	**42.05**	**78.87**	**79.01**	**78.63**	**82.88**
BM-DETR	1. GMR Baseline	4.00	36.32	**75.00**	74.26	**75.00**	71.55	3.65	33.93	78.58	77.52	**78.71**	79.11
	2. GMR + DQ-CGPv3	7.74	38.02	74.43	71.28	74.98	70.69	**7.32**	36.15	77.84	74.69	78.56	**79.64**
	3. GMR + DQ-CGPv3 + GEQ	**7.82**	**43.35**	**75.00**	**75.04**	**75.00**	**72.09**	**7.32**	**40.89**	**78.80**	**78.90**	78.68	79.15
BAM-DETR	1. GMR Baseline	**23.25**	42.92	70.43	66.17	73.68	**78.66**	20.62	40.75	71.79	65.08	74.37	80.99
	2. GMR + DQ-CGPv3	21.52	41.60	71.83	63.77	74.61	77.80	19.57	40.14	72.01	63.90	74.90	80.13
	3. GMR + DQ-CGPv3 + GEQ	22.84	**43.82**	**74.07**	**71.80**	**75.53**	77.41	**20.74**	**41.76**	**77.58**	**73.62**	**78.63**	**83.32**
Sim-DETR	1. GMR Baseline	20.73	37.57	74.69	54.67	75.00	72.97	19.18	35.68	75.72	55.32	74.68	78.87
	2. GMR + DQ-CGPv3	20.63	38.90	75.43	69.57	75.02	75.84	**19.62**	37.24	75.69	70.52	74.82	80.20
	3. GMR + DQ-CGPv3 + GEQ	**21.43**	**44.96**	**75.56**	**75.64**	**75.40**	**76.79**	19.50	**43.05**	**76.12**	**76.90**	**75.55**	**82.41**
UVCOM	1. GMR Baseline	7.12	35.95	**75.00**	**75.06**	75.00	70.03	6.40	34.32	**76.80**	**77.97**	**75.61**	**78.98**
	2. GMR + DQ-CGPv3	17.06	38.75	70.06	56.83	74.47	**76.83**	16.41	37.37	67.20	55.76	72.48	75.32
	3. GMR + DQ-CGPv3 + GEQ	**18.40**	**42.62**	72.28	66.73	**75.16**	76.65	**17.12**	**40.80**	71.45	64.83	73.70	76.29

1. GMR Baseline: Historical unaugmented GMR. 2. GMR + DQ-CGPv3: Plug-and-play DETR Query CGP (beta=0.05, K=16) between Layer 0 & 1.
3. GMR + DQ-CGPv3 + GEQ: Global Event Query head fine-tuned with BCE + native loss (lr=1e-5 / 3e-5, patience=10 on Val G1 mean).
G1-AUC (%): Area Under the Curve of G-mIoU@1 across 101 discrete presence thresholds tau in [0.00, 1.00] (higher is strictly better).
Bold values: Best performance among the available ablation variants within each backbone family.